Department of Electrical and Electronic Engineering
Imperial College London

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EE4-46
Distributed Computation and Networks: a Performance Perspective

Assignment 3: Analysis of Coupled Systems

Submitted by: Pietro Marone (pm2213, CID: 00865920)
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Declaration: I confirm that this submission is my own work. In it, I give references and citations whenever I refer to or use the published, or unpublished, work of others. I am aware that this course is bound by penalties as set out in the College examination offenses policy.
Signed: Pietro Marone



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1. Introduction

In this assignment, we are to analyse a system composed by 2 routers receiving only positive external arrivals, sending only negative customers between each other and sending the remaining customers out of the system. For sake of generality, we will first do our analysis on a system of n routers, obtain generic results, and finally substitute $n = 2$ to obtain results specific to this system.

2. The generic system (n routers)

In the system under analysis, the n queues receive positive customer arrivals from the outside world at Poisson processes of rate $\lambda_i, i = 1, 2, \dots, n$ and have exponentially distributed service rates μ_i. There are no external negative arrivals.

A customer leaving queue i after being serviced will become a negative customer and reach queue j with probability p_{ij}; it will instead depart from the system with probability $(1 - \sum_{j=1}^{n} p_{ij})$; therefore, a customer never leaves a queue to join another as a positive customer.

A negative customer joining queue j from queue i will move a positive customer from queue j to queue i, if queue j is not empty. Otherwise, it will do nothing (so only queue i will be decremented).

Based on the description of the system, we can now write the traffic equations, i.e. equations for positive customer arrivals $\Lambda^+(i)$ and negative customer arrivals $\Lambda^-(i)$ to queue i.

$$\Lambda^+(i) = \lambda_i + \sum_{j=1}^{n} \mu_i p_{ij} q_i q_j \qquad [1]$$

$$\Lambda^-(i) = \sum_{j=1}^{n} q_j \mu_j p_{ji} \qquad [2]$$

Where q_i is the average probability of the queue i to be not empty. We can then write the balance equation, describing the evolution of the queues $\boldsymbol{k}$ over time, where $\boldsymbol{k} = (k_1, k_2, \dots, k_n)$ is a vector containing the queue lengths of each router. First, we write the equation listing all the possible cases for the queues to become $\boldsymbol{k}$ after some amount of time Δt, recalling that a serviced customer can either leave the system or join another queue as a negative customer:

$$\begin{aligned} P(\boldsymbol{k}, t+\Delta t) = {} & P(\boldsymbol{k}, t) P(\text{no departures}) P(\text{no arrivals}) \\ & + \sum_{i=1}^{n} P(\boldsymbol{k}_i^+, t) P(\text{departure from } i) P(\text{leaves system}) P(\text{no arrivals}) \\ & + \sum_{i=1}^{n} P(\boldsymbol{k}_i^-, t) P(\text{no departures}) P(\text{arrival at } i) 1[k_i > 0] \\ & + \sum_{i=1}^{n} P(\boldsymbol{k}_i^+, t) P(\text{departure from } j) P(\text{joins } i) 1[k_j > 0] P(\text{no arrivals}) \\ & + \sum_{i=1}^{n} P(\boldsymbol{k}_i^+, t) P(\text{departure from } i) P(\text{joins } j) 1[k_j = 0] P(\text{no arrivals}) \\ & + P(\text{other events, e.g. multiple departures/arrivals}) \end{aligned} \qquad [3]$$

where $\boldsymbol{k_i^+} = (k_1, k_2, \dots, k_{i-1}, k_i + 1, k_{i+1}, \dots, k_n)$, $\boldsymbol{k_i^-} = (k_1, k_2, \dots, k_{i-1}, k_i - 1, k_{i+1}, \dots, k_n)$ and the function $1[X]$ is defined as $1[X] = \begin{cases} 1 \; if \; X \; is \; true \\ 0 \; if \; X \; is \; false \end{cases}$.

We now need to derive the probabilities for these events, recalling that the time between departures is exponentially distributed with average μ_i and the time between arrivals is exponentially distributed with average λ_i.

As we use a first order approximation of the exponential function, i.e. we neglect all terms $O(\Delta t)$, we can write:

$$\begin{aligned} P[t_{next\ departure\ from\ i} > \Delta t] &= 1 - \mu_i \Delta t, \quad P[t_{next\ departure\ from\ i} < \Delta t] = \mu_i \Delta t \\ P[t_{next\ arrival\ to\ i} > \Delta t] &= 1 - \lambda_i \Delta t, \quad P[t_{next\ arrival\ to\ i} < \Delta t] = \lambda_i \Delta t \end{aligned} \qquad [4]$$

We can now derive equations for the probabilities of departures and arrivals in the interval Δt (again, we neglect all terms $O(\Delta t)$):

$$\begin{aligned} P(no\ departures) &= \prod_{i=1}^{n} (1 - \mu_i \Delta t 1[k_i > 0]) \\ P(no\ arrivals) &= \prod_{i=1}^{n} (1 - \lambda_i \Delta t) \\ P(departure\ from\ i) &= \left[\prod_{j=1, j \neq i}^{n} (1 - \mu_j \Delta t) \right] \mu_i \Delta t = \mu_i \Delta t 1[k_i > 0] \\ P(arrival\ at\ i) &= \left[\prod_{j=1, j \neq i}^{n} (1 - \lambda_j \Delta t) \right] \lambda_i \Delta t = \lambda_i \Delta t \end{aligned} \qquad [5]$$

From the problem statement, we also have these two probabilities for a customer departing from queue i:

$$\begin{aligned} P(joins\ j) &= p_{ij} \\ P(leaves\ system) &= (1 - \sum_{j=1}^{n} p_{ij}) \end{aligned} \qquad [6]$$

Noting that $P(other\ events) = O(\Delta t)$ and can therefore be neglected, we can finally rewrite [3] as follows (note that the condition $1[q_i > 0]$ is not necessary if a term contains $P(\boldsymbol{k_i^+}, t)$, as the fact that the queue i was not empty at time t is already implicit):

$$
\begin{aligned}
P(\boldsymbol{k}, t+\Delta t) = P(\boldsymbol{k}, t) & \prod_{i=1}^{n}(1-\mu_i \Delta t 1[k_i>0]) \prod_{i=1}^{n}(1-\lambda_i \Delta t) \\
& + \sum_{i=1}^{n} P(\boldsymbol{k}_i^+, t)\mu_i \Delta t (1-\sum_{j=1}^{n} p_{ij}) \prod_{i=1}^{n}(1-\lambda_i \Delta t) \\
& + \sum_{i=1}^{n} P(\boldsymbol{k}_i^-, t) \prod_{i=1}^{n}(1-\mu_i \Delta t)\, \lambda_i \Delta t 1[k_i>0] \\
& + \sum_{i=1}^{n} P(\boldsymbol{k}_i^+, t)\mu_j \Delta t p_{ji} 1[k_j>0] \prod_{i=1}^{n}(1-\lambda_i \Delta t) \\
& + \sum_{i=1}^{n} P(\boldsymbol{k}_i^+, t)\mu_i \Delta t p_{ij} 1[k_j=0] \prod_{i=1}^{n}(1-\lambda_i \Delta t)
\end{aligned}
\qquad [7]
$$

Noting that many of the terms resulting from the products are $O(\Delta t)$ and rearranging some of the other terms, we can rewrite the equation as:

$$
\begin{aligned}
P(\boldsymbol{k}, t+\Delta t) = P(\boldsymbol{k}, t) & \left[1-\left(\sum_{i}^{n}(\mu_i \Delta t 1[k_i>0] + \lambda_i \Delta t)\right)\right] \\
& + \sum_{i=1}^{n} P(\boldsymbol{k}_i^+, t)\mu_i \Delta t \left(1-\sum_{j=1}^{n} p_{ij}\right) \\
& + \sum_{i=1}^{n} P(\boldsymbol{k}_i^-, t)\lambda_i \Delta t 1[k_i>0] \\
& + \sum_{i=1}^{n} P(\boldsymbol{k}_i^+, t)\mu_j \Delta t p_{ji} 1[k_j>0] \\
& + \sum_{i=1}^{n} P(\boldsymbol{k}_i^+, t)\mu_i \Delta t p_{ij} 1[k_j=0]
\end{aligned}
\qquad [8]
$$

Rearranging the terms, dividing both sides by Δt and taking the limit as Δt tends to 0 we have:

$$
\begin{aligned}
\lim_{\Delta t \to 0} \frac{P(\boldsymbol{k}, t+\Delta t) - P(\boldsymbol{k}, t)}{\Delta t} = \frac{dP(\boldsymbol{k}, t)}{dt} = & \\
& -P(\boldsymbol{k}) \sum_{i}^{n}(\mu_i 1[k_i>0] + \lambda_i) \\
& + \sum_{i=1}^{n} P(\boldsymbol{k}_i^+)\mu_i \left(1-\sum_{j=1}^{n} p_{ij}\right) \\
& + \sum_{i=1}^{n} P(\boldsymbol{k}_i^-)\lambda_i 1[k_i>0]
\end{aligned}
\qquad [9]
$$

$$+\sum_{i=1}^{n} P(\boldsymbol{k}_i^+)\mu_j p_{ji} 1[k_j > 0]$$
$$+\sum_{i=1}^{n} P(\boldsymbol{k}_i^+)\mu_i p_{ij} 1[k_j = 0]$$

In steady state, $\frac{dP(\boldsymbol{k},t)}{dt} = 0$, and we therefore obtain:

$$P(\boldsymbol{k})\sum_{i}^{n}(\mu_i 1[k_i > 0] + \lambda_i) = \sum_{i=1}^{n} P(\boldsymbol{k}_i^+)\mu_i\left(1 - \sum_{j=1}^{n} p_{ij}\right) + \\ +\sum_{i=1}^{n} P(\boldsymbol{k}_i^-)\lambda_i 1[k_i > 0] + \sum_{i=1}^{n} P(\boldsymbol{k}_i^+)\mu_j p_{ji} 1[k_j > 0] + \sum_{i=1}^{n} P(\boldsymbol{k}_i^+)\mu_i p_{ij} 1[k_j = 0] \quad [10]$$

3. Steady-State Distribution of Queue Length

We now state that the joint probability distribution of the queue length in all queues is given by:

$$P(\boldsymbol{k}) = \prod_{i=1}^{n}(1 - q_i)q_i^{k_i} \quad [11]$$

where

$$q_i = \frac{\Lambda^+(i)}{\mu_i + \Lambda^-(i)} \quad [12]$$

We now need to prove that this is indeed a solution of the balance equation obtained in the previous section.

To do so, we first divide both sides of [10] by $P(\boldsymbol{k})$. Noting that according to [11], $\frac{P(\boldsymbol{k}_i^+)}{P(\boldsymbol{k})} = q_i$ and $\frac{P(\boldsymbol{k}_i^-)}{P(\boldsymbol{k})} = \frac{1}{q_i}$, this yields:

$$\sum_{i=1}^{n}(\lambda_i + \mu_i 1[k_i > 0]) = \sum_{i=1}^{n} q_i\mu_i\left(1 - \sum_{j=1}^{n} p_{ij}\right) + \\ +\sum_{i=1}^{n}\frac{\mu_i}{q_i}1[k_i > 0] + \sum_{i=1}^{n}\sum_{j=1}^{n} q_i\mu_j p_{ji} 1[k_j > 0] + \sum_{i=1}^{n}\sum_{j=1}^{n} q_i\mu_i p_{ij} 1[k_j = 0] \quad [13]$$

Swapping some of the summation indexes and grouping terms yields:

$$\sum_{i=1}^{n}(\lambda_i+\mu_i 1[k_i>0]) = \sum_{i=1}^{n} q_i\mu_i\left(1-\sum_{j=1}^{n} p_{ij}\right)+ \\ +\sum_{i=1}^{n}\frac{1[k_i>0]}{q_i}(\mu_i+q_iq_j\mu_i p_{ij})+\sum_{i=1}^{n}\sum_{j=1}^{n} q_j\mu_j p_{ji}1[k_i=0] \quad [14]$$

Using [1] yields:

$$\sum_{i=1}^{n}(\lambda_i+\mu_i 1[k_i>0]) = \sum_{i=1}^{n} q_i\mu_i\left(1-\sum_{j=1}^{n} p_{ij}\right)+ \\ +\sum_{i=1}^{n}\frac{\Lambda^{+}(i)}{q_i}1[k_i>0]+\sum_{i=1}^{n}\sum_{j=1}^{n} q_j\mu_j p_{ji}1[k_i=0] \quad [15]$$

Using [12] yields:

$$\sum_{i=1}^{n}(\lambda_i+\mu_i 1[k_i>0]) = \sum_{i=1}^{n} q_i\mu_i\left(1-\sum_{j=1}^{n} p_{ij}\right)+ \\ +\sum_{i=1}^{n}\mu_i 1[k_i>0]+\sum_{i=1}^{n}\Lambda^{-}(j)1[k_i>0]+\sum_{i=1}^{n}\sum_{j=1}^{n} q_j\mu_j p_{ji}1[k_i=0] \quad [16]$$

Using [2] yields:

$$\sum_{i=1}^{n}(\lambda_i+\mu_i 1[k_i>0]) = \sum_{i=1}^{n} q_i\mu_i-\sum_{j=1}^{n}\Lambda^{-}(j)+ \\ +\sum_{i=1}^{n}\mu_i 1[k_i>0]+\sum_{i=1}^{n}\Lambda^{-}(j)1[k_i>0]+\sum_{i=1}^{n}\Lambda^{-}(j)1[k_i=0] \quad [17]$$

Cancelling some terms and noting that $1[k_i>0]+1[k_i=0]=1$ yields

$$\sum_{i=1}^{n}\lambda_i = \sum_{i=1}^{n} q_i\mu_i-\sum_{i=1}^{n}\Lambda^{-}(j)+\sum_{i=1}^{n}\Lambda^{-}(j) \quad [18]$$

Cancelling some terms yields:

$$\sum_{i=1}^{n}\lambda_i = \sum_{i=1}^{n} q_i\mu_i \quad [19]$$

In order to prove this, we add $\sum_{i=1}^{n}\sum_{j=1}^{n}\mu_i p_{ij} q_j q_i$ on both sides:

$$\sum_{i=1}^{n}\lambda_i + \sum_{i=1}^{n}\sum_{j=1}^{n}\mu_i p_{ij} q_j q_i = \sum_{i=1}^{n} q_i\mu_i + \sum_{i=1}^{n}\sum_{j=1}^{n}\mu_i p_{ij} q_j q_i \qquad [20]$$

Using [1] and changing some summation indexes yields:

$$\sum_{i=1}^{n}\Lambda_i^+ = \sum_{i=1}^{n} q_i\mu_i + \sum_{i=1}^{n}\sum_{j=1}^{n}\mu_j p_{ji} q_i q_j \qquad [21]$$

Grouping terms on the right-hand side yields:

$$\sum_{i=1}^{n}\Lambda_i^+ = \sum_{i=1}^{n} q_i(\mu_i + \sum_{j=1}^{n}\mu_j p_{ji} q_j) \qquad [22]$$

Using [2] and then [12] finally yields:

$$\sum_{i=1}^{n}\Lambda_i^+ = \sum_{i=1}^{n} q_i(\mu_i + \Lambda^-(i)) = \sum_{i=1}^{n}\Lambda_i^+ \qquad [23]$$

This proves that [11] is the steady-state distribution for queue lengths.

4. Solution for system with $n = 2$

We have thus demonstrated that

$$p(\boldsymbol{k}) = \prod_{i=1}^{n}(1-q_i)q_i^{k_i} \qquad [24]$$

where $q_i = \frac{\Lambda^+(i)}{\mu_i + \Lambda^-(i)}$, $\Lambda^+(i) = \lambda_i + \sum_{j=1}^{n}\mu_i p_{ij} q_i q_j$, $\Lambda^-(i) = \sum_{j=1}^{n} q_j\mu_j p_{ji}$

is the steady-state distribution for queue lengths. For the specific case with $n = 2$, $p_{12} = p_1$, $p_{21} = p_2$, $p_{11} = 0$ and $p_{22} = 0$, this can be written as:

$$p(n,m) = (1-q_1)q_1^n(1-q_2)q_2^m \qquad [25]$$

where $q_1 = \frac{\lambda_1}{\mu_1 + q_2\mu_2 p_2 - \mu_1 p_1 q_2}$, $q_2 = \frac{\lambda_2}{\mu_2 + q_1\mu_1 p_1 - \mu_2 p_2 q_1}$

5. Conclusion

In this assignment we analysed a simple network of routers in order to obtain equations for the joint probability distribution of the queue lengths at all routers. First, we derived the generic distribution for n routers, and then we obtained the distribution $p(n, m)$ for the case of 2 routers only.

6. References

(1) Gelenbe, E. (1991) Product-form queuing networks with negative and positive customers. Journal of Applied Probability, Vol. 28, No. 3

(2) Gelenbe, E. and Mitriani, I. (1980) Analysis and Synthesis of Computer Systems. Academic Press, London

(3) Gelenbe, E. and Pujolle, G. (1986) Introduction to Queuing Networks. Wiley, Chichester